



Workloads, Scalability and QoS Considerations in CMP Platforms

Presenter – Don Newell

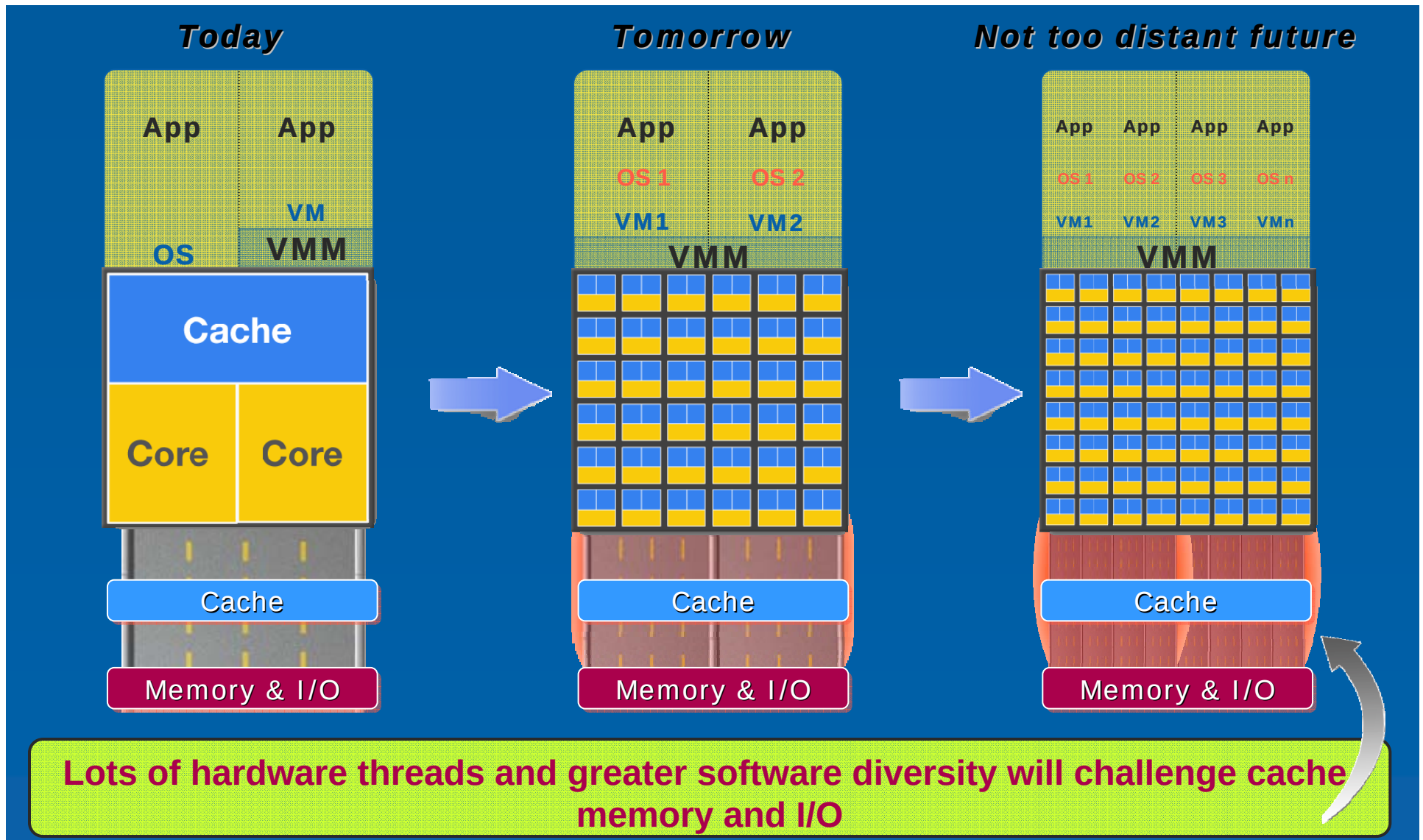
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Agenda

- **Trends and research context**
- **Evolving Workload Scenarios**
- **Platform Scalability**
- **Platform QoS**

Trends

- Cores, Threads, and Virtual Machines

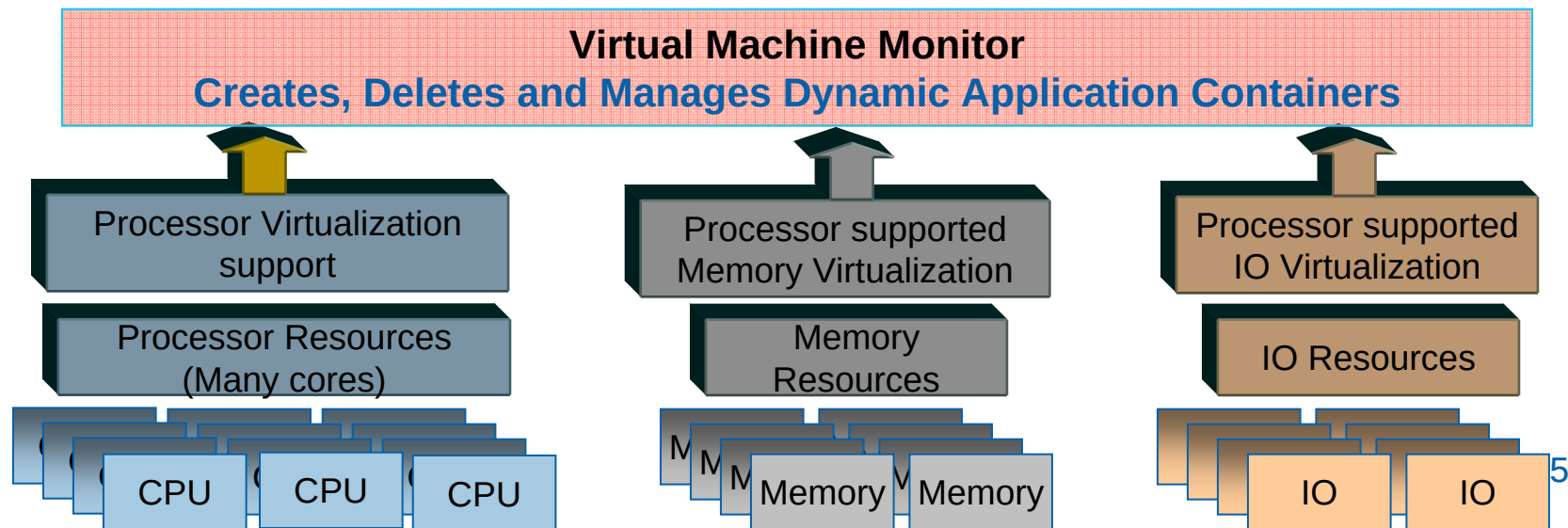
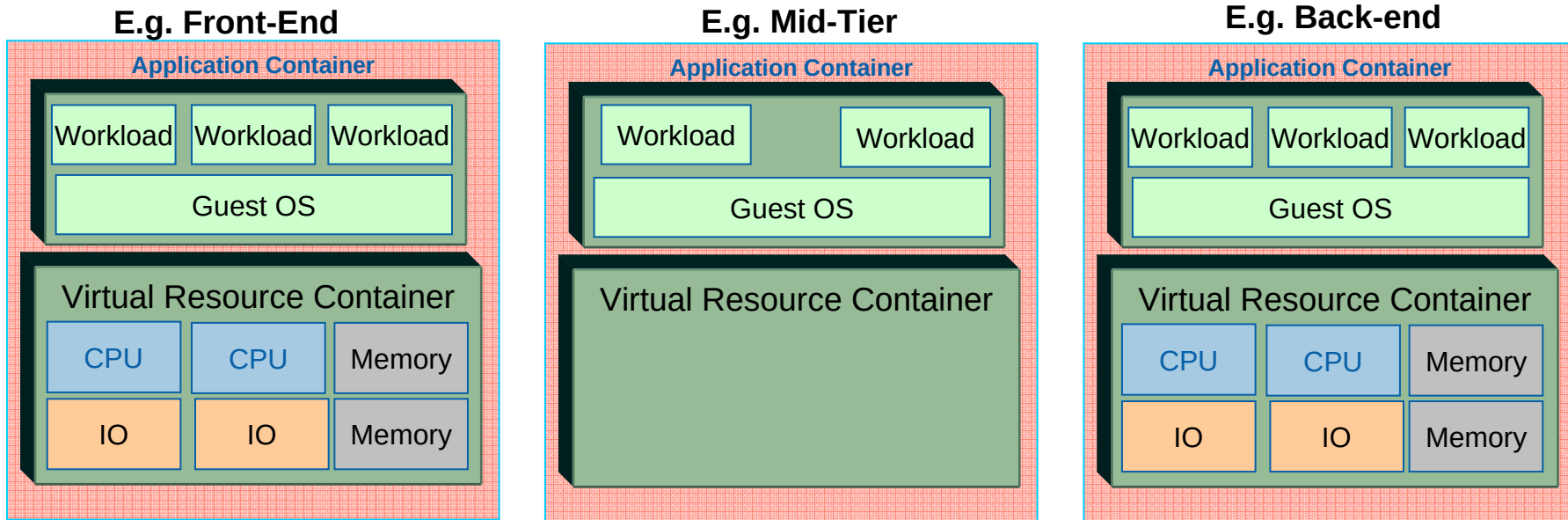


Trends - Discussion Points

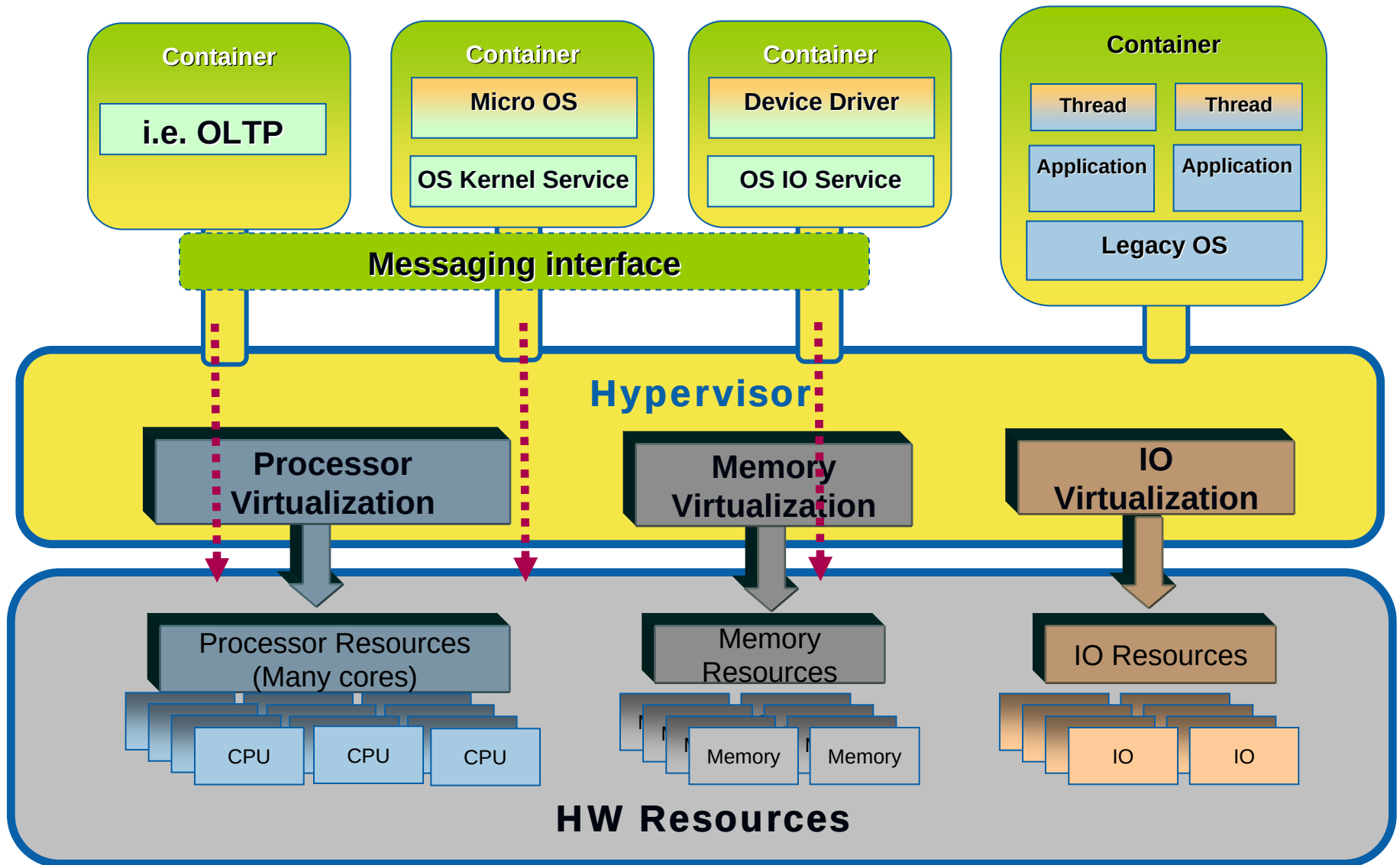
Goal: Scaling up to 100's of logical processors on a single CPU die

- Scaling hardware threads means provisioning and re-architecting other platform resources
- Scaling hardware threads means learning to share – QoS revisited

Trends - Pool of Virtual Resources

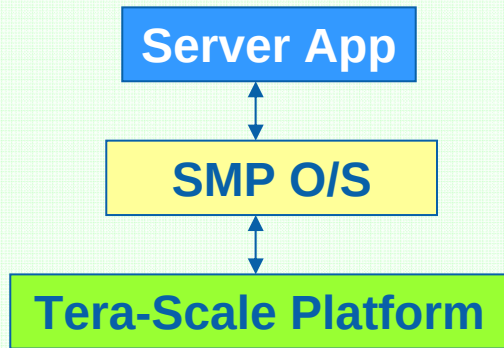


Trends - Decomposed OS



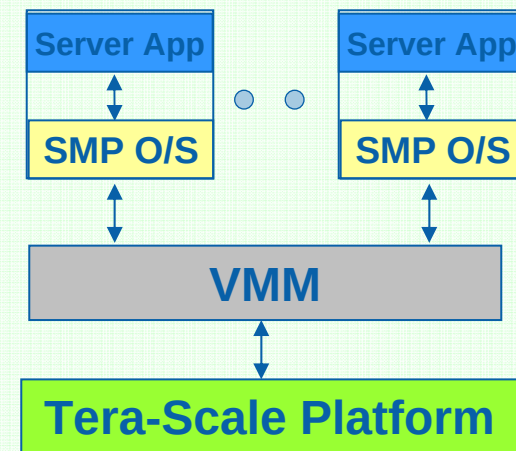
Platform Scaling

Tera-Scale Workload Scenarios



Highly Multi-threaded Server Workloads

OLTP, J2EE App Servers, E-commerce, ERP, Search, etc
(TPC-C, SPECjbb2005, SAP, SPECjappserver2004, etc)



Virtualized Server Environments

(Workload Consolidation, Datacenter-on-chip Scenarios)

Cache / Memory Considerations:

Performance => Overall Throughput

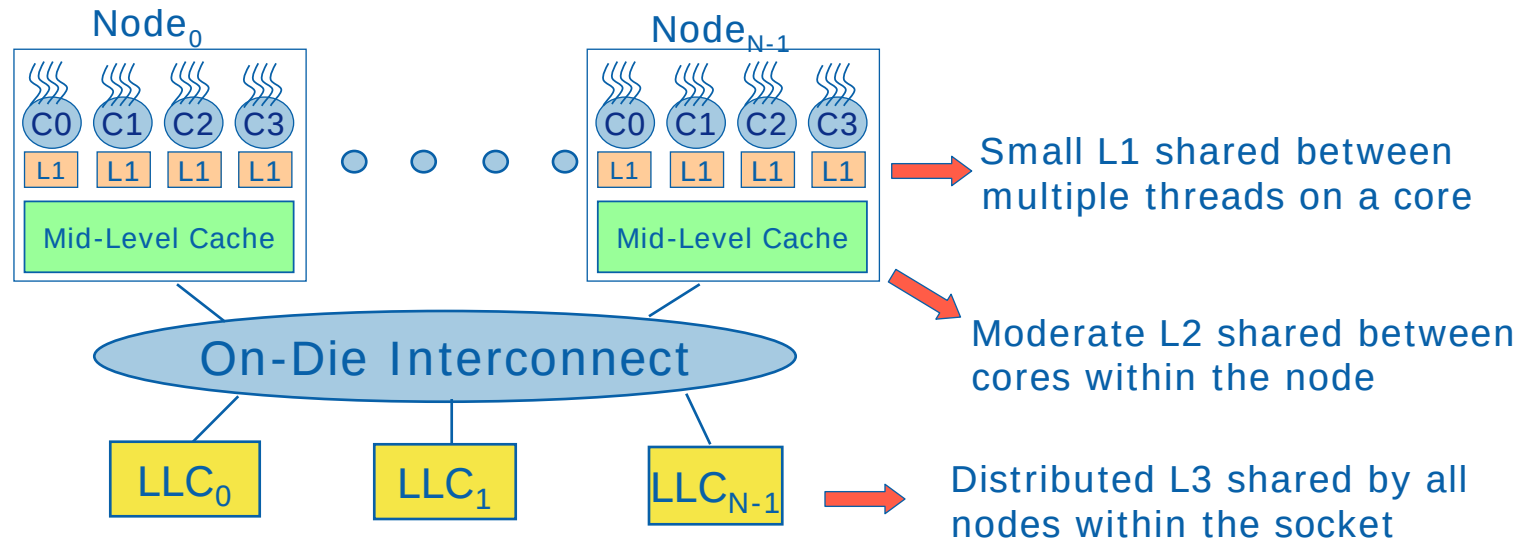
Scalability => Headroom

Quality of Service => Performance Isolation

Let's begin with single server app.....

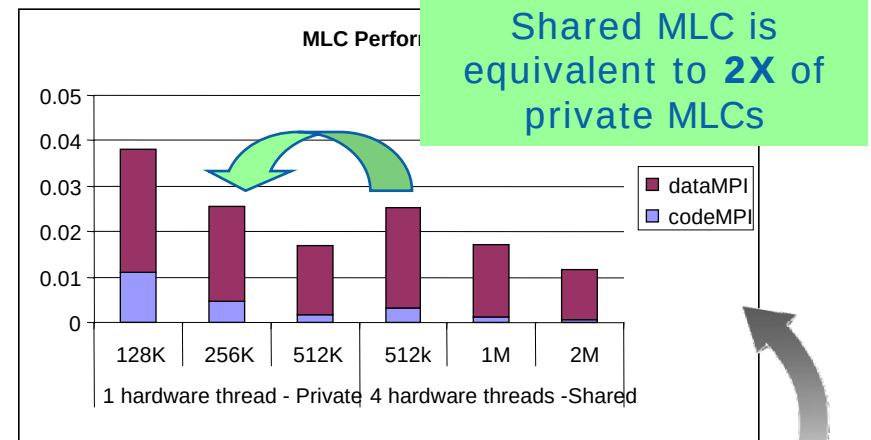
And follow up with the consolidated scenario.....

Tera-Scale Cache Hierarchy Design



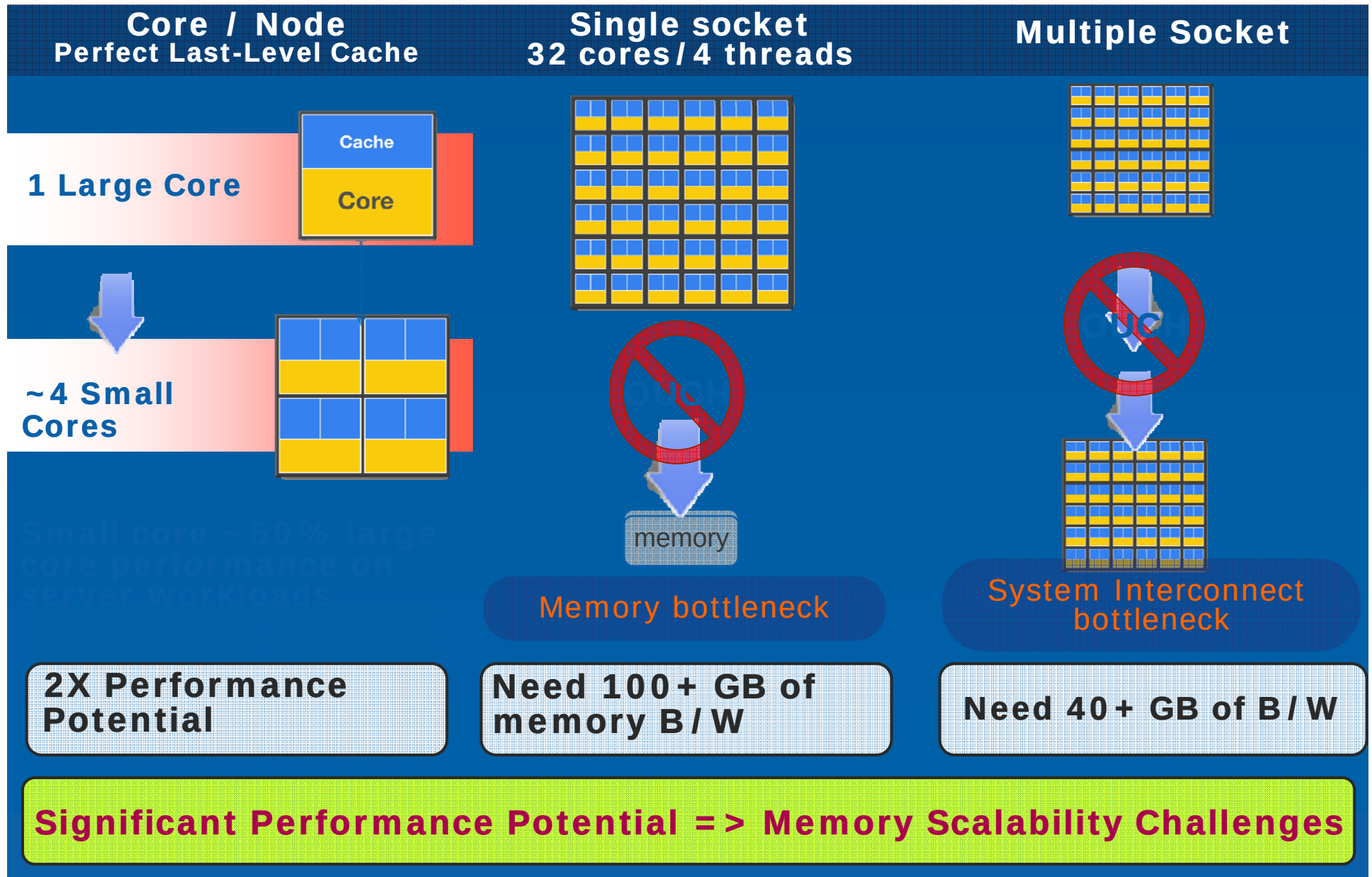
Hierarchical Sharing Benefits

- Reduces Replication
- Increases Effective Cache Size
- Better localized communication
- Reduced interconnect pressure



Hierarchical sharing increases caching effectiveness significantly

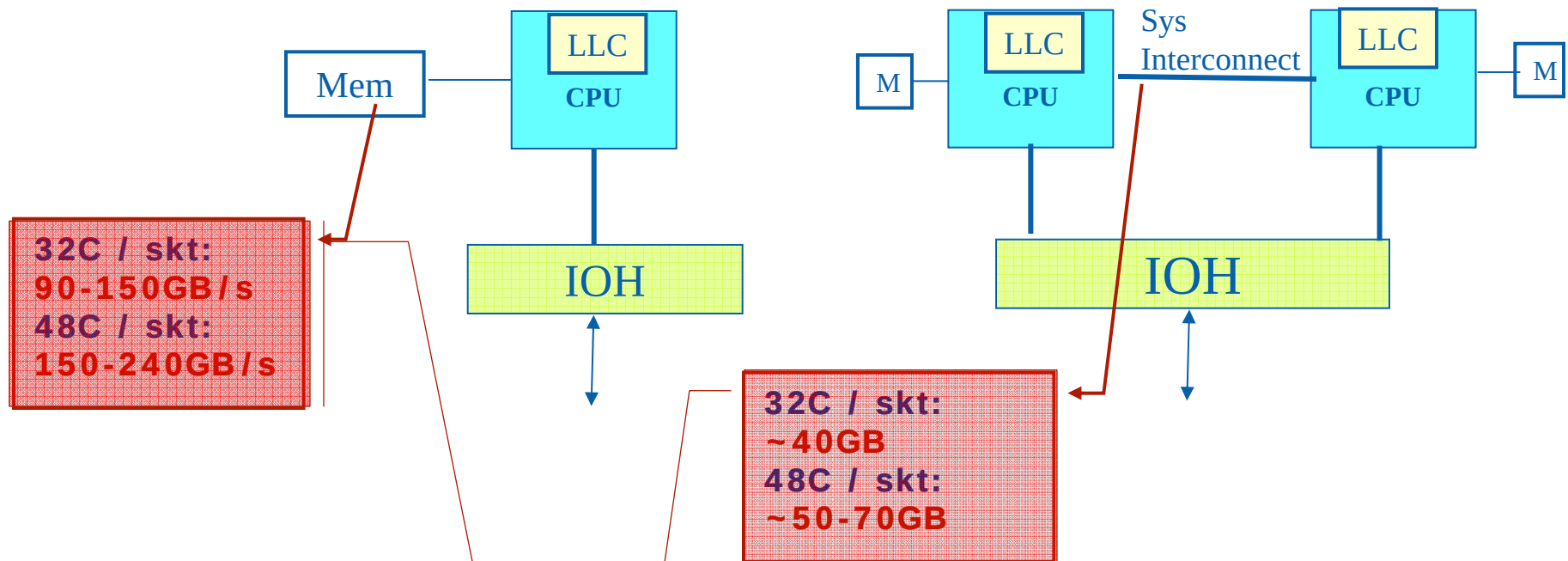
OLTP Tera-Scale Case Study



Scalability Issues

Tera-scale Headroom requirements

- Start with 32 cores in 1st gen; Maybe grow to 48 cores in next gen?
- How much memory & interconnect bandwidth will we need?



Scaling Issues Just Get Worse

On-Socket DRAM Caches (For Memory Scalability)

Enable Large Capacity L4s

- Low Latency
- High Bandwidth

Technologies

- 3D Stacking
- Multi-chip Packages (MCP)

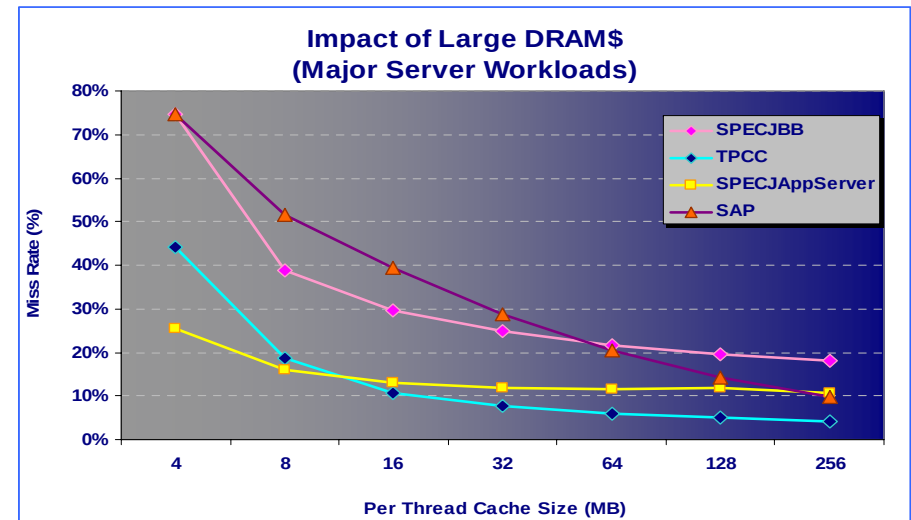
Benefits

- Significant reduction in miss rate
- Avoids bandwidth wall

3D stack



MCP



QoS and Performance Management

Background for QoS Discussion



Trends

- Increasing Core Count for Performance
- Increasing Workload Diversity
 - *Multi-Workload Scenarios in the Client*
 - *Virtualization and Consolidation in the Server*
 - *Heterogeneous Architectures for Graphics*



Observations

- Multi-core enables simultaneous execution of multiple workloads
- But not all Workloads are equal -- users do have preferences

How well does the user-preferred application run?
Should platforms optimize for the user-preferred application?

Resource Management

Capitalist

- No management of resources
- If you can generate more requests, you will use more resources
- Grab as you will
- *E.g. All of today's policies*

Communist/Fair

- Fair distribution of resources
- Give equal share of resources to all executing threads
- Does not necessarily guarantee equal performance
- *E.g. Partitioning resources for fairness and isolation*

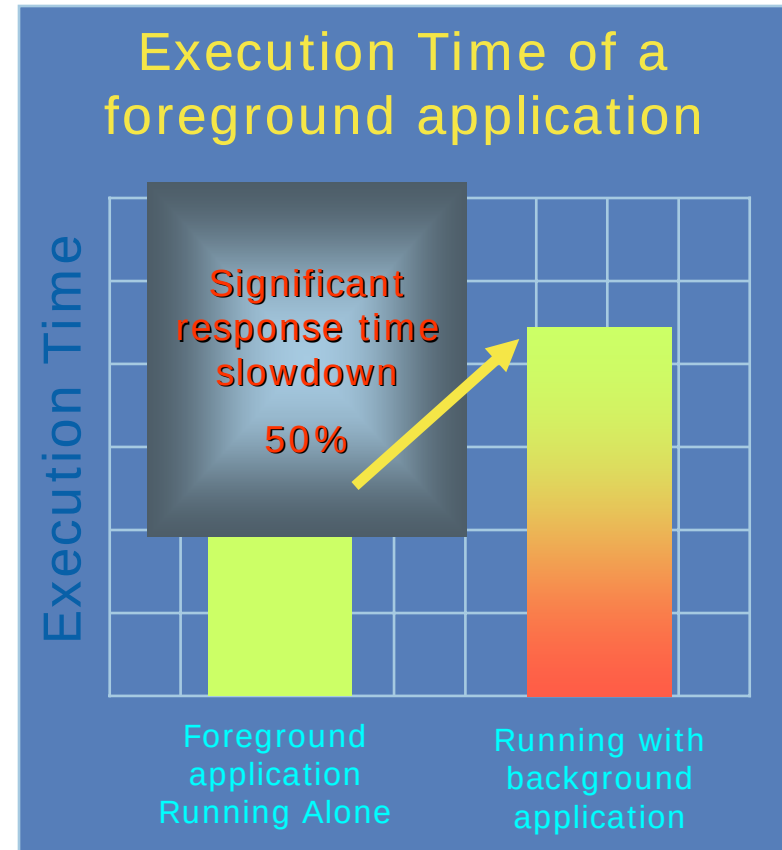
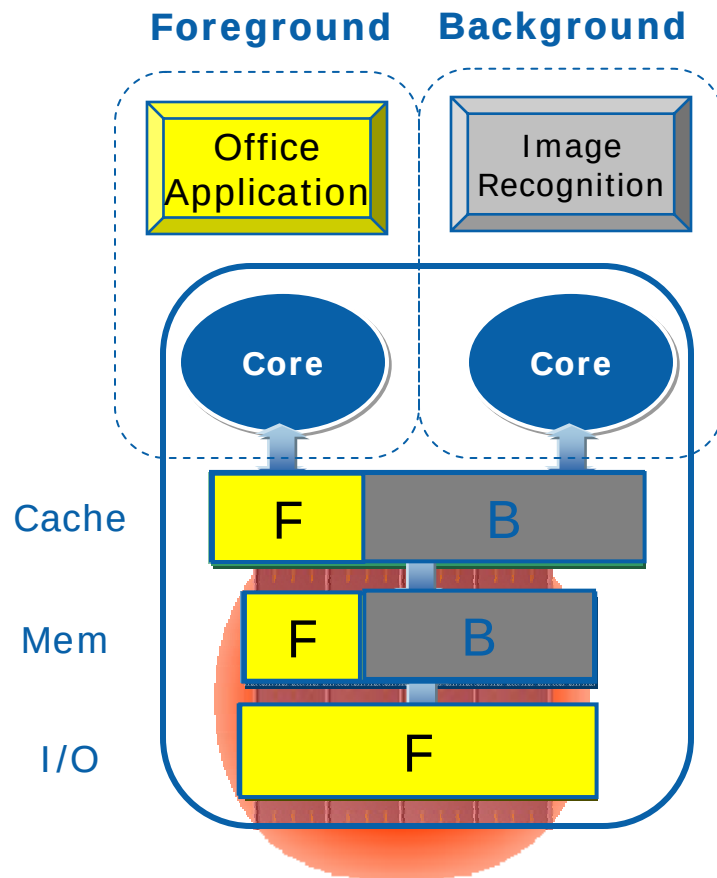
Elitist

- Focus on individual efficiency
- Provide more performance and resources to the VIP
- Limited resources to non-VIP
- *E.g. Service Level Agreements, Foreground / Background*

Utilitarian

- Focus on overall efficiency
- Give more resource to those that need it the most, less to others
- *E.g. Cache-friendly vs. Unfriendly, resource-aware scheduling*

The Multi-Workload Problem



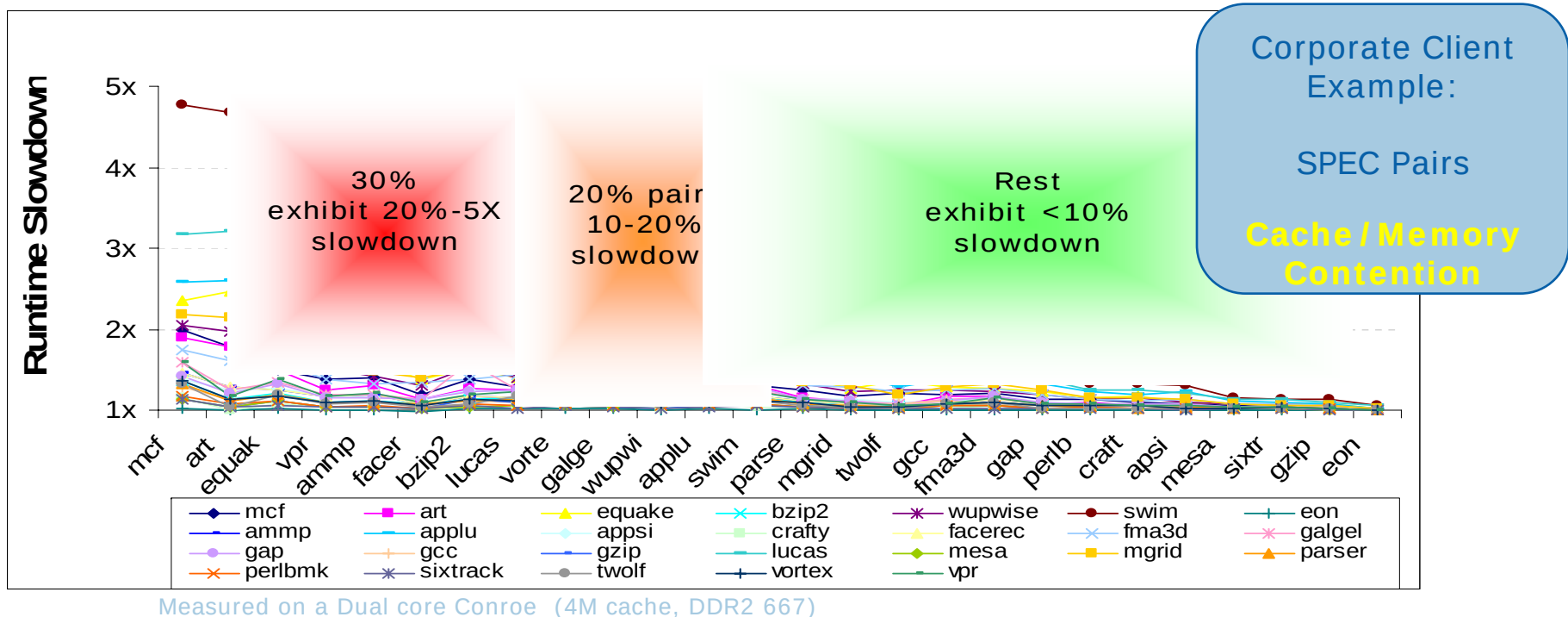
Conroe core 2 Duo Measurements

Platform does not distinguish in resource allocation

Preferred (foreground) application can suffer significant slow down

Platform QoS can improve user-preferred application performance

Contention - Client side examples

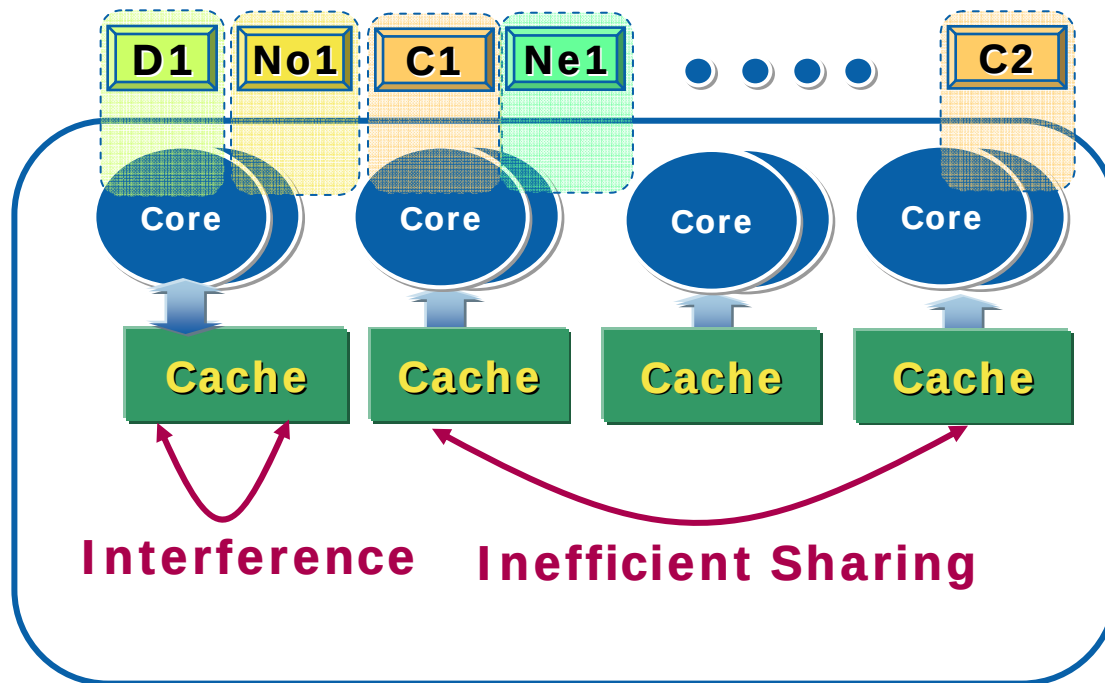


Similar data collected for server applications

Resource contention will impair the performance of important apps
(Performance Differentiation)

Application Behavior & Overall Performance

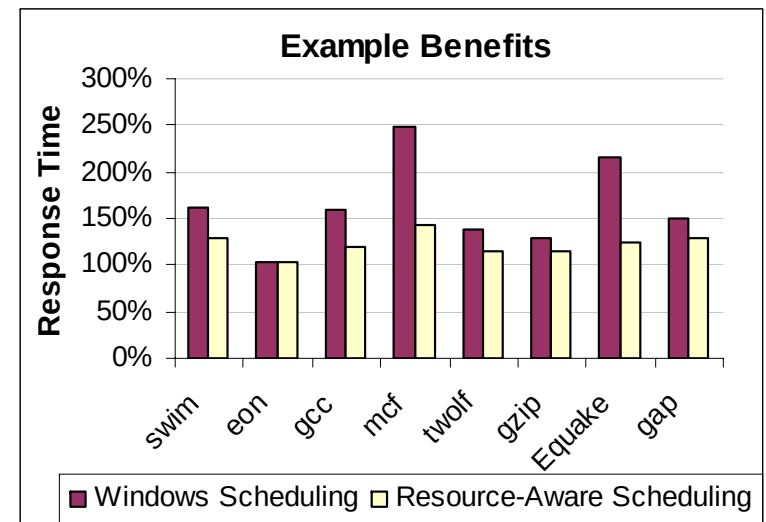
Destructive (streaming) Normal (typical) Constructive (co-op threads) Neutral (little data)



Many Heterogeneous Applications

Potential to improve overall performance

Monitor resource usage and group/partition accordingly

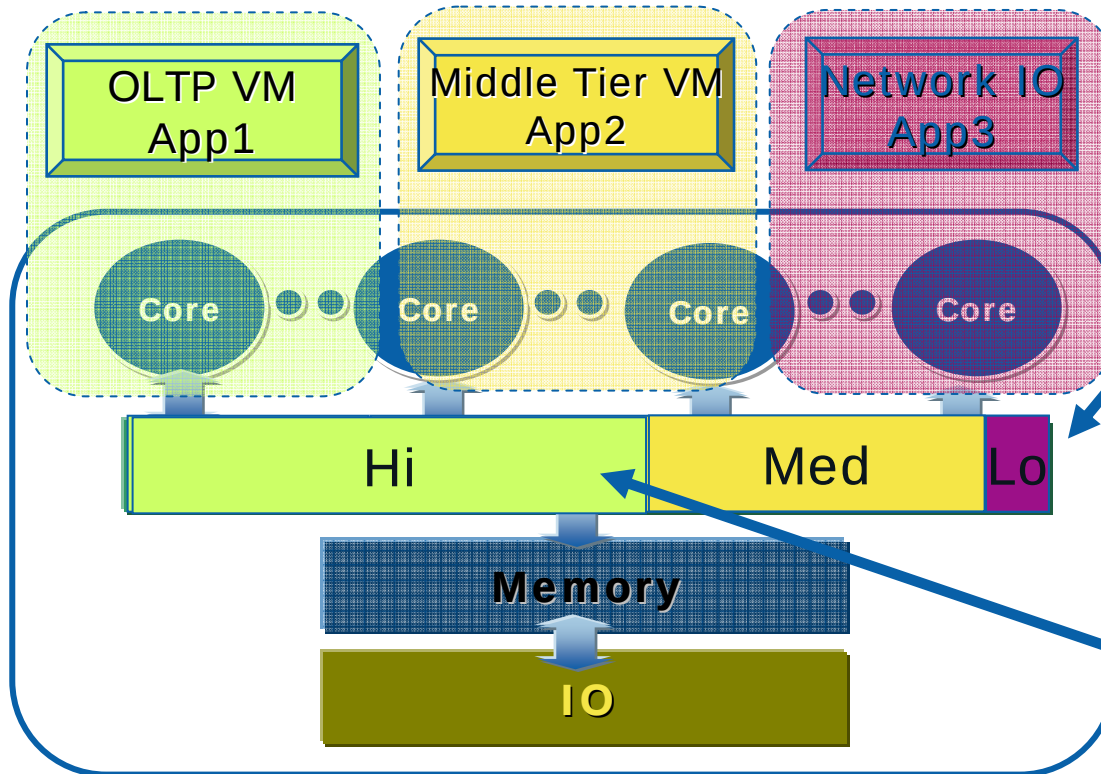


Clovertown (8Core / 8App) Experiments
(used destructive, normal and neutral apps)

Managing resource contention can improve overall throughput too
(Performance Management)

Service Level Agreements in the Enterprise

Server Consolidation



Many Different Virtual Machines

Utility SLA

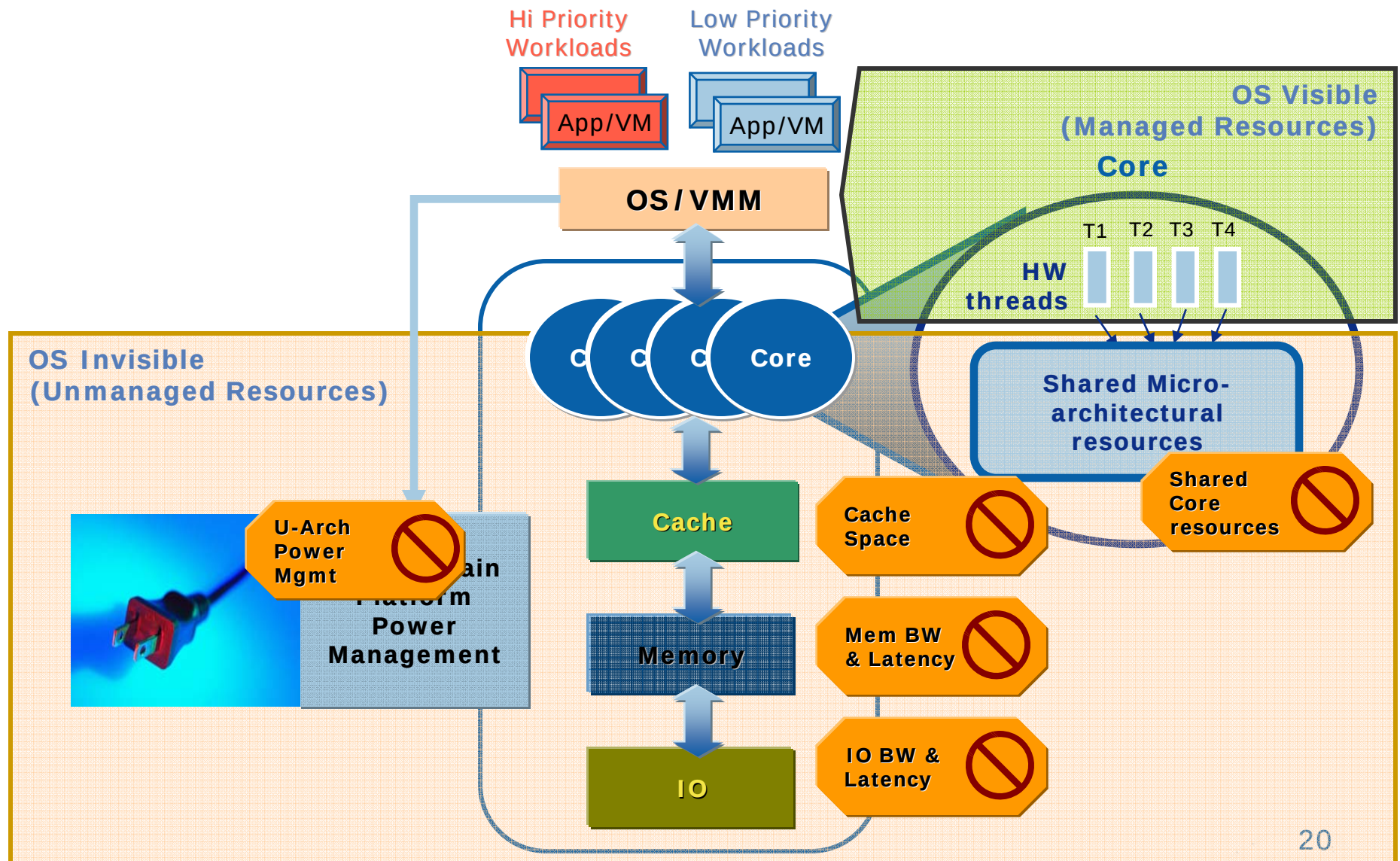
Disparate resource usage can cause performance isolation concerns

SLA management

Prioritized resource allocation can help address performance isolation and SLA

Disparate resource usage and contention hurts SLAs
(Need for Performance Isolation and SLA enforcement)

Shared Platform Resources



Problem Summary

- **CMP → Many heterogeneous threads, apps, VMs**

Not all applications are equal

- **Users have preferences**
 - End-users (client) want to treat foreground preferentially
 - End-users (server) want service level differentiation (SLA) or isolation
- **Applications use resources differently**
 - Destructive vs. Constructive vs. Neutral Threads
 - No performance management to protect from bad behavior

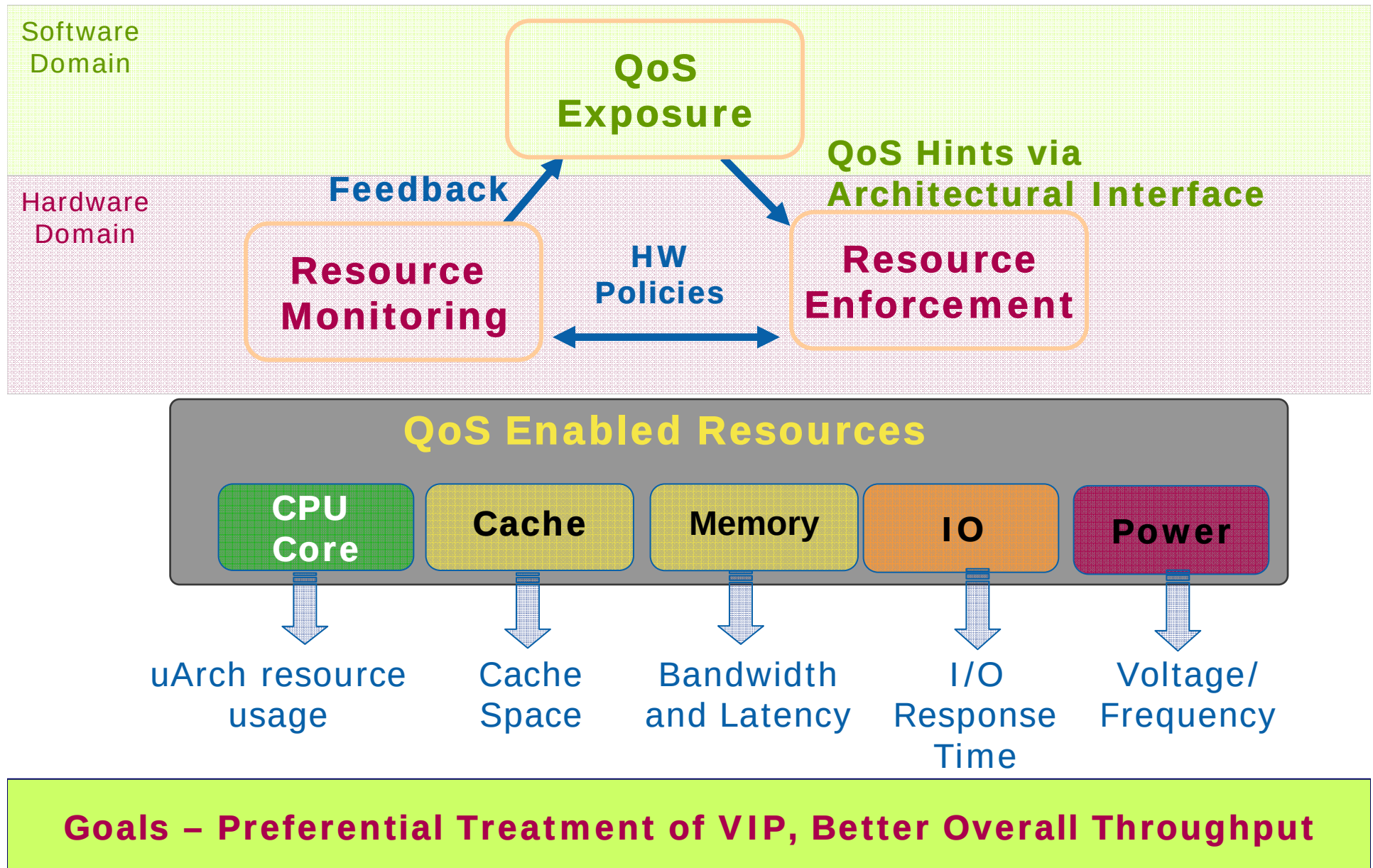
• Priority-based OS scheduling no longer sufficient

- With more cores, OS will allow high and low priority applications to run simultaneously and contend for resources
- Low priority applications will steal platform resources from high priority apps → loss in performance & user experience

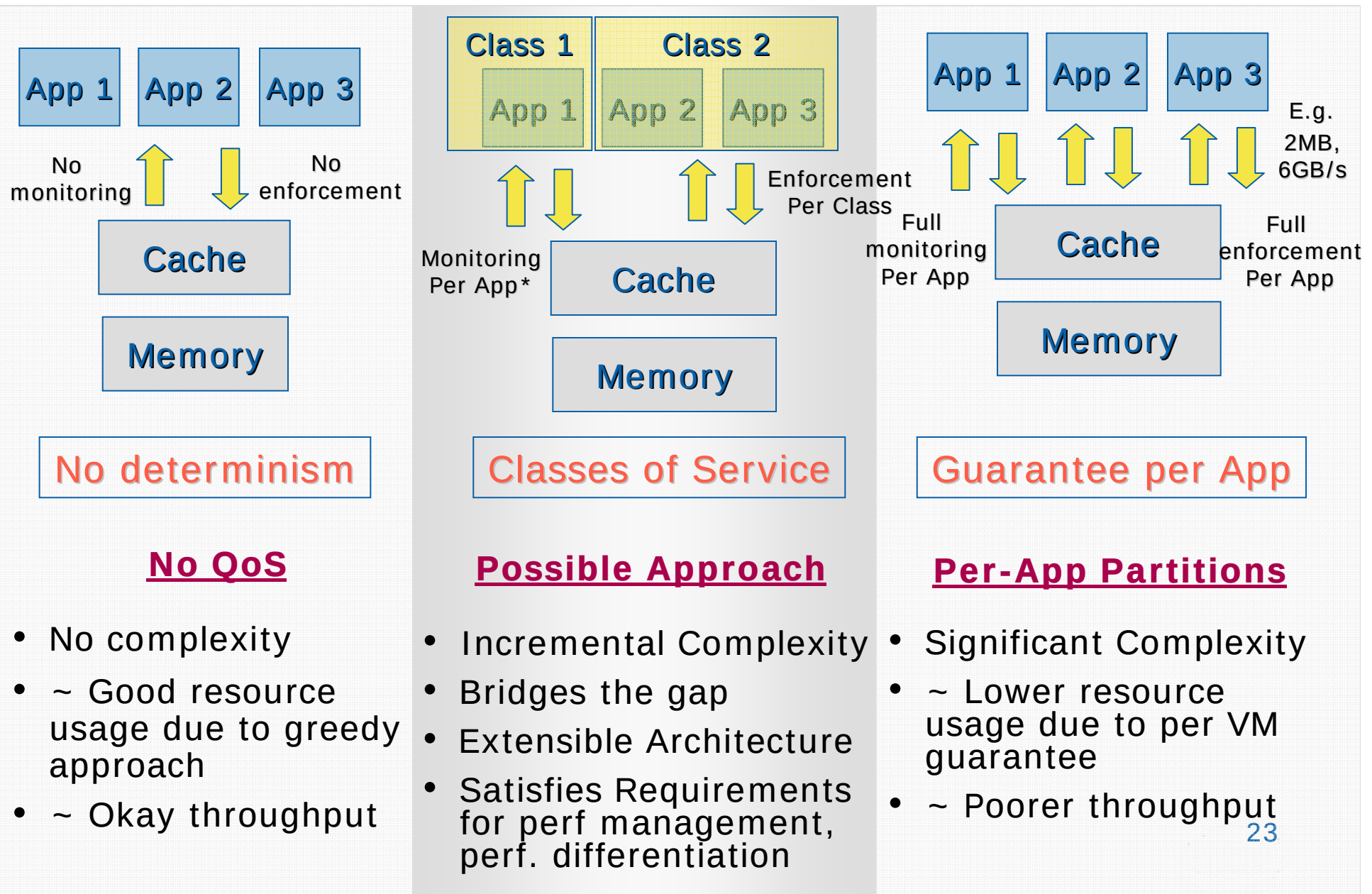
• Platform has no support for application differentiation

- Platform has no knowledge of preferences or resource usage
- Platform has no support for fine-grained tracking of many shared resources that have significant performance value

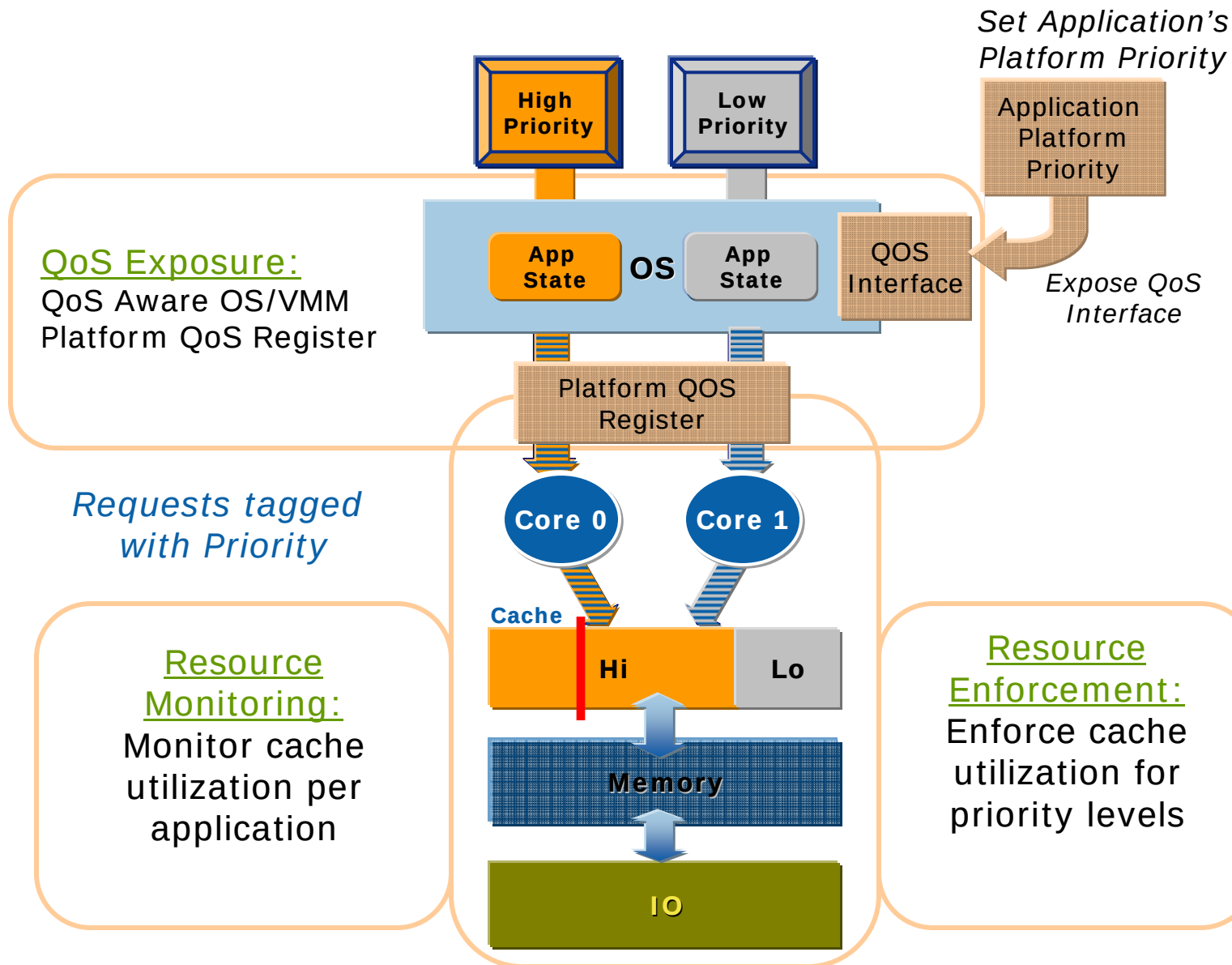
Platform QoS



Visible QoS Spectrum (Cache/Memory)

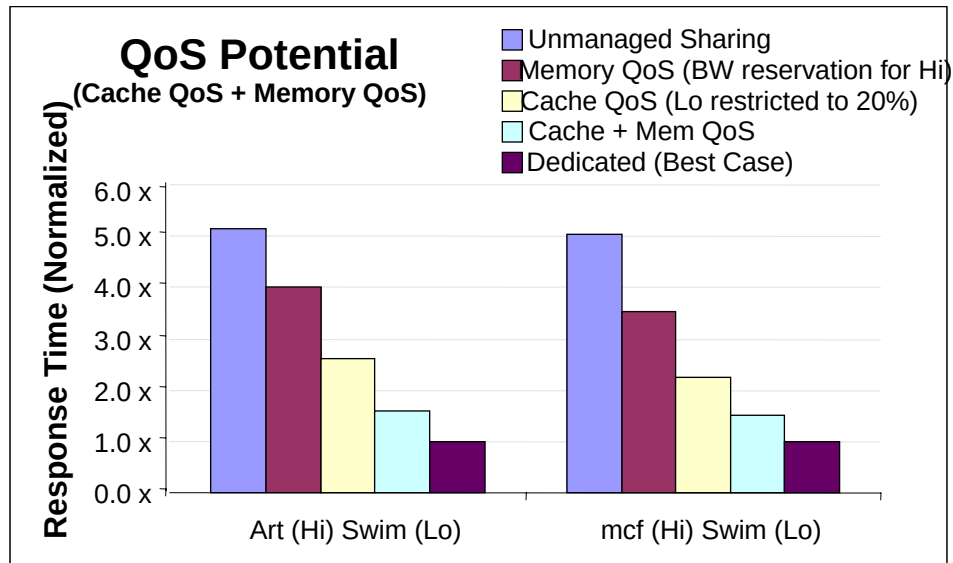


QoS Aware Architecture - Cache



Cache / Memory QoS Benefits

Client SPEC Case Study

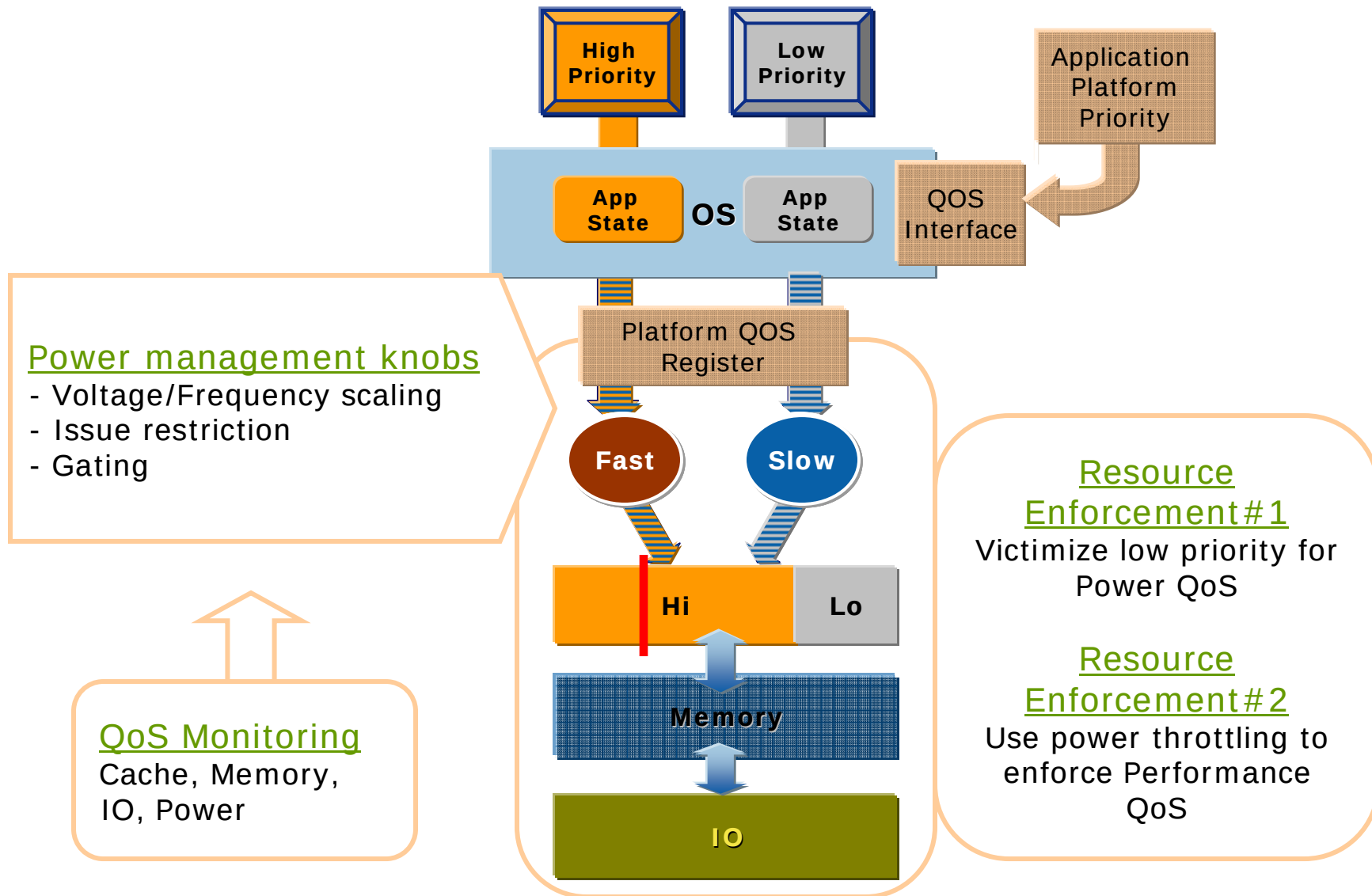


Response Time -- Lower is better

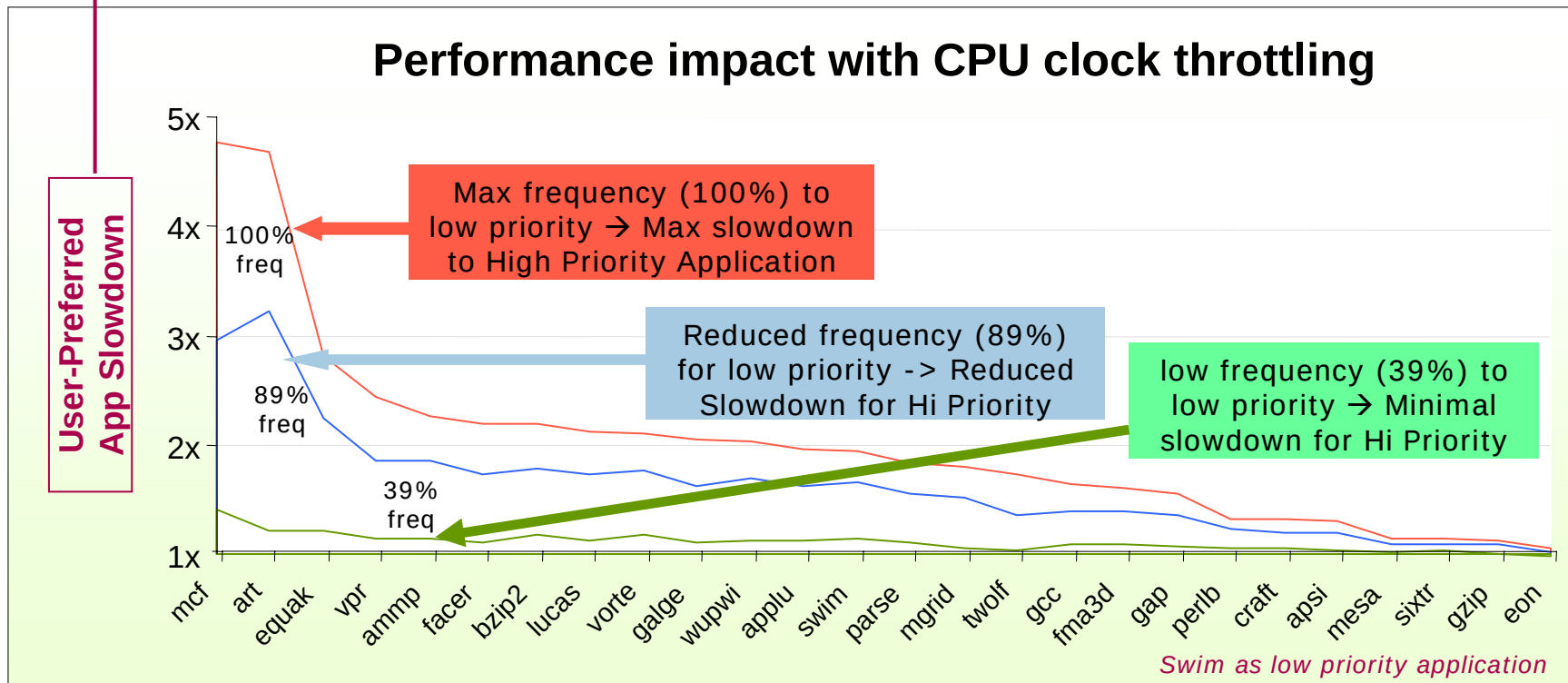
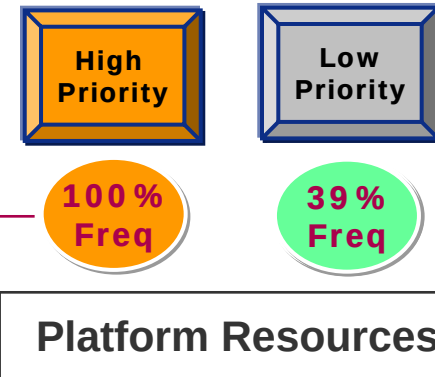
Based on Measurements, Simulations and Analytical Projections

Significant Benefits of Cache / Memory QoS

QoS Aware Architecture: Power



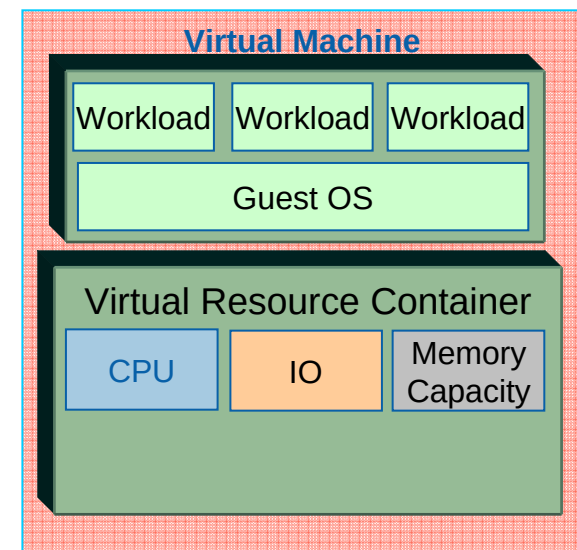
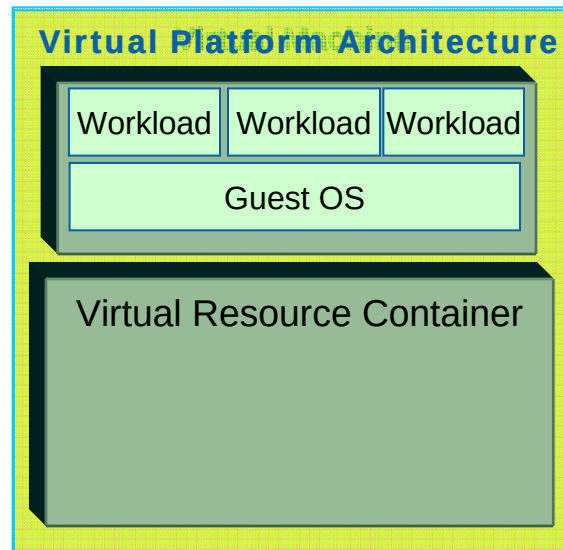
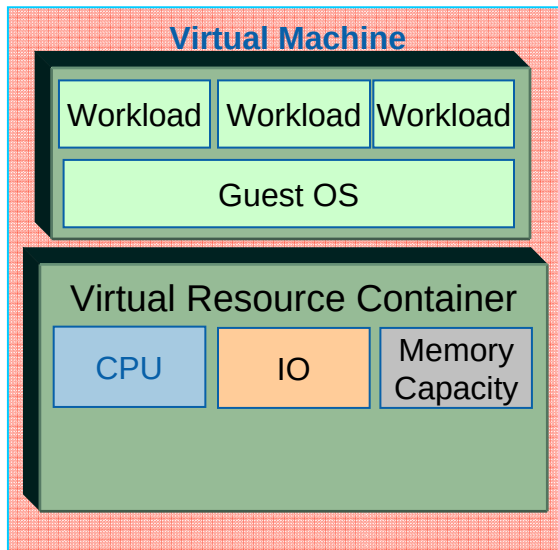
Power QoS Benefits



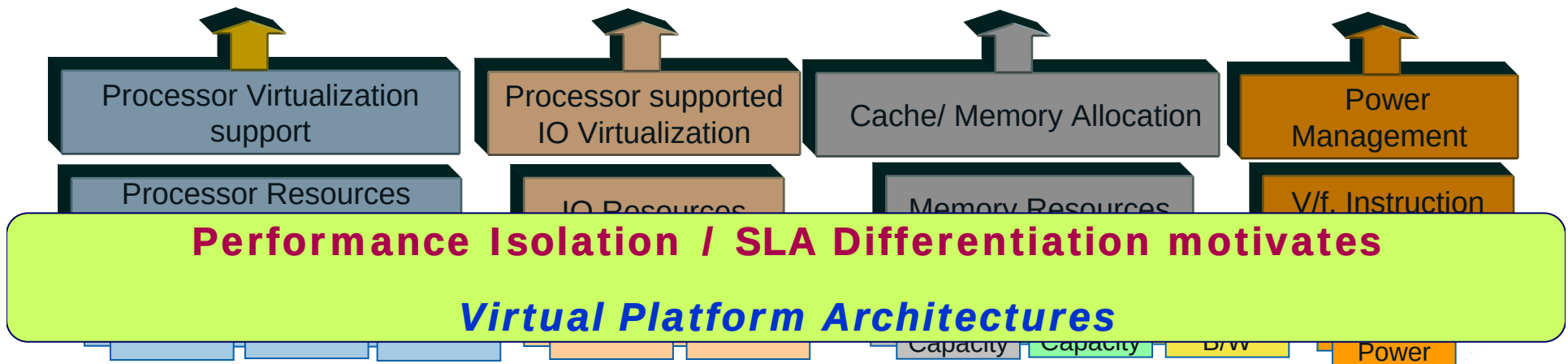
Improves performance of the user-preferred application

Virtualization: From VMs to VPAs

(Managing Transparent Resources)



Virtual Machine Monitor manages CPU, IO and memory capacity
Virtual Platform Architectures manages performance with PQoS



Summary

Large-scale CMP is going to happen

- Lots of work to be done to identify and remove platform and architectural limitations preventing applications and execution environments from scaling up to 100's of logical processors on a single CPU die
- Scalability concerns can be addressed
 - Hierarchy of Shared Caches
 - Large DRAM caches
- QoS concerns can be addressed
 - Dynamic Cache Allocation (Cache QoS)
 - Dynamic Power Management (Power QoS)
- Smart performance management requires more visibility be available to the execution environment
 - Resource utilization counters for schedulers, etc.